Fashion and Textiles

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Modeling and development of continuous textile structures utilizing 3D printing

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Abstract

This study explores the development of continuous textile structures utilizing Thermoplastic Polyurethane (TPU) through advanced 3D printing techniques—namely, Material Extrusion (MEX) and Conveyor Material Extrusion (CMEX). The research aims to address limitations in conventional textile manufacturing by introducing digitally fabricated textile architectures that offer design flexibility, production efficiency, and sustainability. Various TPU filaments were comparatively evaluated to determine optimal material properties, including thermal stability, tensile strength, and dyeability. Three textile geometries—plain weave, flat knit, and spacer fabric—were modeled and fabricated under optimized printing conditions. By adjusting parameters such as nozzle angle, line thickness, and inter-yarn gap, the study achieved structurally stable and visually consistent textile forms. CMEX printing, which utilizes an inclined nozzle and continuous conveyor bed, demonstrated superior capability for producing uninterrupted textile lengths, overcoming the spatial constraints of traditional MEX platforms. Surface morphology analyses and mechanical tests revealed that spacer fabrics exhibited the highest tensile and tear strengths, while flat knit structures maintained superior flexibility. Dyeing experiments confirmed that disperse dyes produced the most stable coloration, with good wash and moderate dry-cleaning fastness. This research contributes to the growing body of literature on additive manufacturing in textiles by offering practical insights into the feasibility of TPU-based printed textiles for apparel use. The findings highlight the potential of CMEX 3D printing to revolutionize textile production, enabling sustainable, on-demand, and customizable garment manufacturing.

Keywords: 3D printing, AM(additive manufacturing), TPU(Thermoplastic Polyurethane), Textile structure, Textile modeling, Textile

Introduction

Sustainability has consistently emerged as a critical concern in the fashion industry, impacting both humanity and the environment not only in the present but also in an uncertain future. It encompasses environmental concepts focused on survival and ecological efficiency (Na, Lee, & Na, 2014). Sustainability challenges the traditional clothing production methods, which involve complex processes with long lead times and extensive waste generation (Dimitrov et al., 2006; Wang & Chen, 2014). In response to these issues, the concept of Zero Waste fashion has been introduced. The term "Zero Waste" is

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a combination of "Zero," meaning none, and "Waste," meaning trash, signifying the elimination of waste (National Institute of Korean Language, n.d). In fashion design, Zero Waste involves minimizing waste during the manufacturing process, such as cutting and sewing fabrics, thereby reducing economic and environmental losses (Yim, Kim, & Beom, 2018).

In this context, Three-Dimensional Printing (3D printing) technology, which creates three-dimensional structures by printing based on digital image data using thermoplastic or curable materials, has gained attention for its ability to shorten lead times, minimize waste, and enable personalized designs (Chakraborty & Biswas, 2020). Additionally, 3D printing offers advantages in product design by integrating cost efficiency, a wide selection of materials, and innovative designs. It also provides various conveniences to designers by allowing the computer-aided design (CAD) of complex and flexible structures, which can be easily designed, sliced, and downloaded via the internet (Chakraborty, 2016). Due to its production efficiency, 3D printing has traditionally been utilized in manufacturing sectors such as aerospace, automotive, and shipbuilding for producing components like jet engine nozzles, automotive tire molds, and hinges and brackets for vehicle bonnets (Industry News, 2025a, 2025b; Vanaei & Zirak, 2024). However, with advancements in technology and the emergence of diverse printing materials, 3D printing is increasingly being explored in fields such as healthcare, construction, footwear, and apparel industries. Applications now extend to the production of personalized medical prosthetics like hearing aids and dentures, construction of houses and bridges, manufacturing of shoe midsoles and heels, accessories like necklaces, rings, sunglasses, and bags, as well as clothing items such as wigs, bikinis, and dresses (CNET, n.d; Industry News, 2025c; Jeong, 2016; Paľovčík et al., 2025; Science Times, 2025).

Previous studies in the fashion domain utilizing 3D printing primarily focused on producing accessories, trimmings, or applying unit-based structural designs to garments due to the limited print bed size of conventional 3D printers (Kim, 2016; Lee, 2015; Lee, Hwang, & Kim, 2015; Lee & Kim, 2015; Lee, Kim, & Hwang, 2016; Lee & Lee, 2016; Paľovčík et al., 2025; Vanderploeg, Lee, & Mamp, 2017). These unit-based structures require multiple prints of the smallest component parts that make up the garment, which must then be sewn or assembled (Lee, 2019), presenting a significant drawback. Moreover, the structural designs typically necessitate the use of rigid plastic materials such as Acrylonitrile Butadiene Styrene (ABS) and Polylactic Acid (PLA). However, ABS and PLA are inherently hard and brittle, lacking the elasticity and flexibility required for textile applications, making them unsuitable for use in clothing (Jeon et al., 2013). Additionally, while there is a need for objective evaluations of printed textiles for their applicability in garments, existing research has predominantly focused on design aspects rather than foundational studies on print conditions and material property assessments (Tolmaç & İşmal, 2024).

To address these issues, there is a need for research utilizing 3D printers with improved bed limitations and the ability to print continuous textile structures. Furthermore, the printing materials should possess enhanced flexibility, making them suitable for textile applications, such as Thermoplastic Polyurethane (TPU). However, TPU presents its

own challenges, as it comes in various hardness levels. Higher hardness TPUs exhibit excellent abrasion resistance but offer poor wear comfort, whereas lower hardness TPUs provide superior comfort but lack durability (Jeon et al., 2013). Therefore, it is crucial to compare TPUs of different hardness levels to select appropriate TPU filaments for apparel materials. Additionally, objective evaluations of print conditions and material properties are necessary to assess the applicability of printed textiles in garments.

This study aims to utilize Conveyor Material Extrusion (CMEX) 3D printing technology, which features a conveyor belt-based print bed, to produce continuous textile structures. The research seeks to explore the feasibility of printing these structures by optimizing print conditions. Consequently, the study intends to provide foundational data for evaluating the potential application of printed textiles as garment materials through objective fiber property assessments. The significance of this research lies in its use of conveyor belt-based 3D printing technology to produce flexible TPU-based textiles and its provision of objective evaluation results regarding print conditions and material properties, thereby assessing the applicability of the printed structures in apparel.

However, due to the limited bed size of conventional Material Extrusion (MEX) 3D printing systems, this study introduces a continuous conveyor belt-based system (CMEX) to enable uninterrupted production of textile structures. Unlike MEX, where material is deposited perpendicularly onto the bed, CMEX employs an inclined nozzle angle that significantly influences layer adhesion, geometry, and overall structural integrity**.** While previous research in other fields has explored conveyor-belt-based 3D printing, these studies have primarily focused on non-textile applications using small-scale conveyor setups unsuitable for garment-scale fabrication. Moreover, no prior work has demonstrated the direct production of interlaced textile structures for apparel using conveyor-based 3D printing. Even attempts to mimic textile-like forms have relied on upright, layer-by-layer deposition, forfeiting the inherent advantages of continuous fabrication and failing to achieve truly woven, interconnected structures. To address these limitations, this study employs a CMEX system with an expanded conveyor platform to enable the continuous production of flexible textile forms while systematically identifying the minimum printable thickness achievable with TPU filaments, acknowledging the material constraints of additive manufacturing. By optimizing print parameters within these boundaries, the research establishes the feasibility of continuously producing textile-like structures with improved flexibility and integrity. Furthermore, it investigates and proposes the most stable printing angles and conditions for CMEX, providing a logical and empirically supported rationale for selecting optimal deposition angles essential for producing structurally sound continuous textile forms, supported by comparative analyses of the surface morphology of MEX- and CMEX-printed samples.

In this study, the primary objectives are threefold. Firstly, to identify the optimal Thermoplastic Polyurethane (TPU) filament suitable for use as an apparel material, a continuous transverse helical coil structure will be modeled to emulate a textile framework without causing interference in the longitudinal direction. This involves evaluating the physical properties of four different TPU filaments to determine the most appropriate filament for this application.

Secondly, to provide foundational data for the production of continuous textile structures, existing textile architectures will be modeled. Specifically, three types of textile structures will be selected and modeled to establish basic data on textile printing. These structures include the plain weave, which is the most fundamental fabric type; a loop-connected structure; and spacer fabric, the basic structure among weaves. These three structures inherently differ in their three-dimensionality, thickness, and layer configuration, meaning that even when using the same TPU filament and printing equipment, the resulting print quality and form can vary significantly due to differences in geometric design. To achieve the best reproduction of each structure, this study therefore includes the determination of optimal printing angles and gap sizes tailored to each design. The modeled textile structures will then be printed using both Material Extrusion (MEX) and Conveyor Material Extrusion (CMEX) 3D printing techniques to observe and analyze their surface morphology (Fig. 1).

Thirdly, the study will assess the dyeability, wash durability, and dry-cleaning durability of the textile structures printed with TPU filament. Specimens will be selected based on the most stable surface morphology angles for each textile structure. These selected specimens will undergo evaluations of tensile strength, bending strength, tear strength, and elongation to determine the suitability of the three proposed textile structures for use in apparel.

Based on these objectives, the research aims to develop continuous textile structures using TPU filament through 3D printing. By evaluating the dyeability, durability, and appearance of the developed textile structures, the study seeks to present objective physical property assessments to determine their potential application as apparel materials. This comprehensive evaluation will provide essential data for assessing the feasibility and practicality of incorporating 3D-printed TPU-based textiles into the fashion industry, addressing both functional performance and aesthetic considerations.

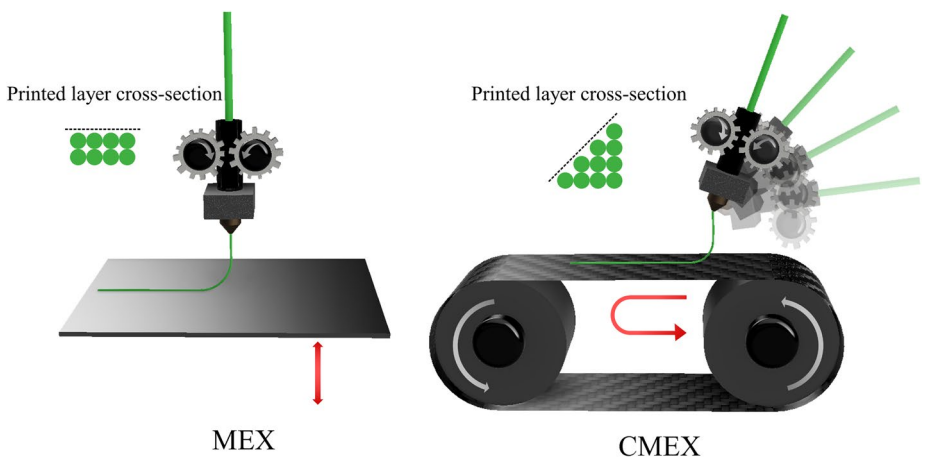


Fig. 1 Schematic diagrams of 3D printing MEX(Material Extrusion) and CMEX(Conveyor Material Extrusion) for filament yarn output

Experimental

3D modeling and slicing

To set the printing parameters for textile structures, modeling of plain weave, plain knit, and spacer fabric structures was conducted using Fusion360 software (Autodesk, Inc., USA).

G-code files were generated from the modeled structures using Cubicreator 3.6.8 software for the CUBICON Single Plus-320C printer and Blackbelt Cura 3.6.0 software for the Blackbelt 3D printer.

3D printing conditions

Two types of 3D printers were used in this study. The printing conditions for each printer were based on the basic output conditions for TPU filament, and are shown in Table 1.

3D printing

Two types of 3D printers were used in this study: a CUBICON Single Plus-320C (Cubicon, Korea) for MEX (Material Extrusion) printing and a Blackbelt 3D printer (Blackbelt 3D B.V., Netherlands) for CMEX (Continuous Material Extrusion) printing. Both printers were used in their original configurations without any modifications to nozzle diameter or bed material.

In MEX, the printed objects are deposited vertically (90°) onto a fixed platform, regardless of the printing strategy. Consequently, the size of the printed object is highly limited, typically ranging between 10 and 20 cm. This limitation makes continuous manufacturing impractical, rendering large-scale production similar to conventional textile processes unfeasible. When applying this method to the fabrication of wearable materials, it is impossible to print a structure large enough to cover the human body in a single process. Therefore, the textile must be printed in smaller modular components that are assembled and sewn together to form a complete garment, which poses a significant drawback.

As an alternative to overcome the limitations of the fixed-bed size, CMEX 3D printing technology was developed for continuous and scalable production. Unlike conventional MEX, which deposits material in horizontal or vertical directions, CMEX utilizes an inclined nozzle (15°, 25°, 35°, 45°) to extrude material onto a conveyor belt platform. This approach allows continuous deposition and enables large-scale manufacturing. The conditions according to the printing type are compared in Table 2.

Table 1 Two types of 3D printing slicing and output conditions

Printing type	MEX	CMEX
Slicing software	Cubicreator 3.6.8	Blackbelt Cura 3.6.0
Infill pattern	Line	
Infill density (%)	100	
Nozzle temperature (°C)	230	
Bed temperature (°C)	65	
Printing speed (mm/s)	30	
Layer thickness (mm)	0.2	

Table 2 Main printing condition of two types 3D printing

Printing type	MEX	CMEX
AM process	Material Extrusion	Continuous Material Extrusion
Nozzle-bed degree (°)	90	15, 25, 35, 45
Nozzle diameter (mm)	0.4	

Table 3 Characteristics of filaments

Filament	Sample code	Product name	Manufacturer	Shore hardness	Thickness (mm)
TPU	E-TPU	eFlex	eSun	87A	1.75
	A-TPU	95A	eSun	95A	
	C-TPU	Cheetah	Ninjatek	95A	
	D-TPU	Armadillo	Ninjatek	75D	

Parts printed with the MEX method are labeled with a nozzle angle of 0°, where 0° represents the standard, non-adjustable vertical nozzle used in traditional printers.

Material

Four thermoplastic polyurethane (TPU) filaments were used. The specifications and sample codes for each filament are presented in Table 3.

Characterization

Filament analysis

The composition of the four TPU filaments was analyzed using Fourier Transform Infrared (FTIR) Spectroscopy (Nicolet iS50, Thermo Fisher Scientific, USA) with attenuated total reflectance (ATR). Measurements were recorded from 7800 to 100 cm^{-1} at a scan rate of 65 scans/sec and a resolution of 0.09 cm^{-1}.

Thermal degradation behavior of the TPU filaments was analyzed using a thermogravimetric analyzer (TGA, Q500, TA Instruments, USA) under a nitrogen atmosphere at a heating rate of 20 °C/min, from 20 to 800 °C.

The thermal properties of the TPU filaments were analyzed using a differential scanning calorimeter (DSC 4000, PerkinElmer, USA) under a nitrogen atmosphere at a heating rate of 20 °C/min, from 30 to 350 °C.

Tensile strength was measured using a universal testing machine (Instron®-3343, Instron, USA) dp 1kN loadcell. For the comparison of physical properties when the plaents were printed into a structure, the specimen was evaluated using a coil-type specimen (Kim & Lee, 2021) produced in previous studies.

3D printing textile analysis

The surface morphology of the textile structures was examined using a stereomicroscope (SMZ-171, Motic, Hong Kong, China) and a digital microscope camera (Moticam S3, Motic, Hong Kong, China). Surface and side views of the plain weave, plain knit, and spacer fabric structures were analyzed.

The tensile strength and elongation of the four TPU filaments were measured using a universal testing machine (Instron 3343, Instron, USA) with a gauge length of 50 mm and crosshead speed of 200 mm/min. The tensile properties of the printed textile structures were evaluated using a universal testing machine (QM100SE, QMSYS, Korea) The evaluation was performed in accordance with Korean industrial standards, KS K 0521 (Textiles—Tensile properties of fabrics—Determination of maximum force and elongation at maximum force).

The flexural strength of the textile structures was tested under optimal printing conditions using a universal testing machine (QM100SE, QMSYS, Korea) The evaluation was performed in accordance with Korean industrial standards, KS M ISO 14125 (Fibre-reinforced plastic composites- Determination of flexural properties).

Tear strength of the optimal printed textile structures was assessed using an Elmendorf tear tester (QM-137, QMSYS, Korea), with a capacity of 6400 gf was evaluated performed in accordance with using the strip method according to the Korean industrial standard in accordance with KS K ISO 13937-1(Textiles-Tear properties of fabrics-Part 1: Determination of tear force using ballistic pendulum method(Elmendorf)).

Stiffness of the optimal textile structures was evaluated using a heart-loop tester (ASA-154, Korea Science Co., Korea), The evaluation was performed in accordance with Korean industrial standards, following KS K 0538 (Textiles—Methods for determination of stiffness of fabrics—Heart loop method).

Dyeing analysis

For colorfastness testing, a multifiber adjacent fabric (Multifiber adjacent fabric TV Style 41, James Heal, UK) was used. Various dyes were employed to examine dye uptake: disperse dye (Blue SEGLS, JIDYES, Korea), basic dye (Blue GRL 300, JIDYES, Korea), and acid dye (Blue E-R, JIDYES, Korea). Acetic acid (Samchun Chemicals Co., Korea) was used as a mordant for basic and acid dyes, sodium sulfite (Samchun Chemicals Co., Korea) as a leveling agent for acid dye, and rod oil (Artnara, Korea) as a dispersant for disperse dye. For post-treatment of basic and acid dyed samples, tannic acid (Samchun Chemicals Co., Korea) was used. AATCC 1993 Standard Reference Detergent WOB (AATCC, USA) was used as a washing detergent, and perchloroethylene (Dajung Chemicals & Metals Co., Korea) as a dry-cleaning solvent. All reagents were of analytical grade.

The dyeing process included various dye types and conditions. Disperse dyeing was performed at a liquor ratio of 30:1 with 3% (o.w.f.) dye and 1.5 g/L rod oil at 40 °C. Basic dyeing involved acetic acid as mordant and was conducted at 80 °C, while acid dyeing used sodium sulfite at 80 °C, with post-treatment using tannic acid. Dye exhaustion was measured by UV–Vis spectrophotometry, and K/S values were calculated using the Kubelka–Munk equation.

Colour fastness to washing of the optimal textile structures was evaluated using a Launder O meter (Launder O meter, Korea science, Co., Korea). The evaluation was performed in accordance with Korean industrial standards, following KS K ISO 105 (Textiles-Tests for colour fastness-Part C01: Colour fastness to washing: Test 1) and KS K ISO 105-D01 (Textiles—Tests for colour fastness—Part D01: Colour fastness to dry-cleaning using perchloroethylene solvent).

Experimental design

The experimental design is summarized schematically in Fig. 2. This figure provides an overview of the sequential stages of material selection, 3D printing under optimized conditions, and evaluation of the printed textile structures.

Results and Discussion

Filament property evaluation

We conducted a multi-modal characterization of four commercial TPU filaments to establish a processing window and to screen materials for subsequent 3D printing experiments. Figure. 3 compiles the spectroscopic (FT-IR), thermoanalytic (TGA and DSC), and mechanical (tensile) results, with cross-references in the text below.

The composition of the four TPU filaments was analyzed by FT-IR spectroscopy. The spectra for each filament exhibited characteristic polyurethane peaks, including a C=O urethane band at 1740–1750 cm^{-1}, an –H urethane band at 1520–1540 cm^{-1}, an N–H urea band at 3330–3400 cm^{-1}, and a CH_2 stretching vibration near 2860 cm^{-1}. The consistent presence of these features across all four filaments indicates no significant compositional differences. However, variations in the relative intensity of the CH_2 stretching near 2860 cm^{-1} suggest differences in alkyl-segment content, which may influence mechanical flexibility and printability (Hepburn, 1992).

Thermogravimetric analysis (TGA) was performed to evaluate thermal stability. Rapid thermal decomposition began near 200 °C, and substantial mass loss continued to and beyond 300 °C, indicating char formation. Despite this onset, extrusion during FFF 3D printing is typically conducted at 220–240 °C because degradation kinetics are sufficiently slow at these temperatures and nozzle residence times are short, minimizing

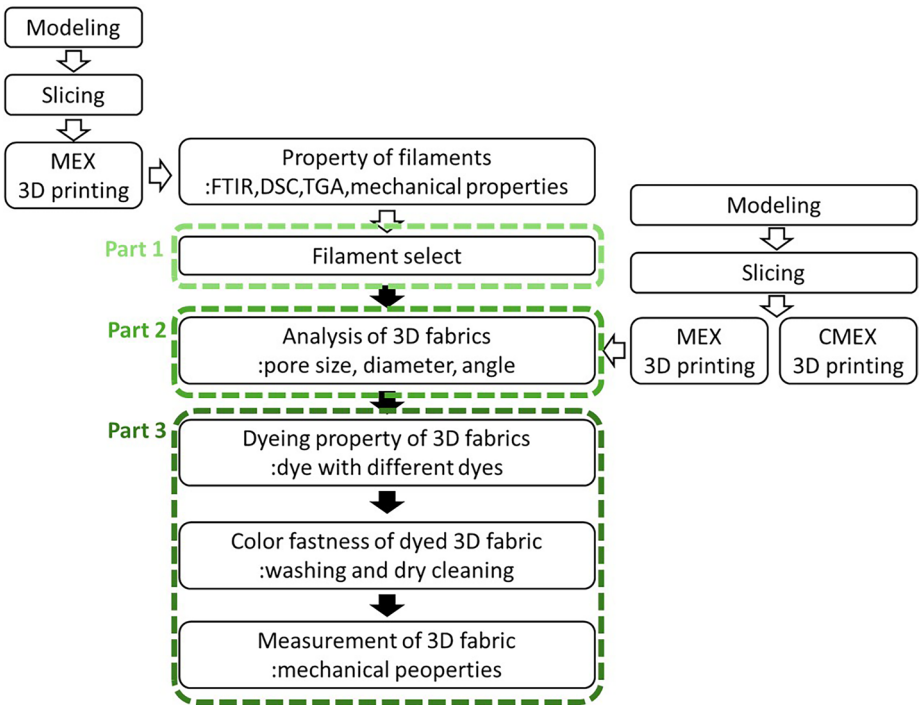


Fig. 2 Schematic process of experimental design

decomposition. In this study, extrusion was carried out at 230 °C; temperatures below 220 °C did not permit smooth flow through the nozzle.

Differential scanning calorimetry (DSC) was used to establish suitable nozzle and bed temperatures. All four filaments displayed thermal behavior typical of amorphous polymers, with similar endothermic and exothermic responses and no distinct crystallization or melting transitions. The softening range was approximately 80–180 °C, and typical processing temperatures for extrusion were 180–230 °C. Notably, D-TPU showed a subtle endothermic feature around 180 °C, consistent with softening, while gradual thermal decomposition was observed above 300 °C. Accordingly, the bed was set below the softening range at 80 °C, and the nozzle was set to 230 °C to optimize printing.

Tensile testing showed elongation exceeding 200% for all filaments, indicating excellent ductility. Among them, A-TPU exhibited the highest force at break, demonstrating superior tensile performance; therefore, A-TPU was selected for subsequent 3D printing.

Modeling of textile structures and surface analysis of the 3D-printed textile structures

Plain weave

The modeling of plain weave textile structures was conducted by varying the line thickness (0.6 mm and 0.8 mm) and the gap size (0.6 mm, 1.0 mm, 1.5 mm, 1.8 mm, 2.0 mm, 4.0 mm, 6.0 mm, 8.0 mm, and 10.0 mm). The modeling images are presented in Table 4. As illustrated in the front view and side view of Table 4, the height of the sketch lines was set to the sum of the line thickness and half the line thickness radius to prevent overlapping of the lines. Additionally, the zigzag sketch lines composed of up and down sections were modeled by gently rounding the angles using the Fillet tool.

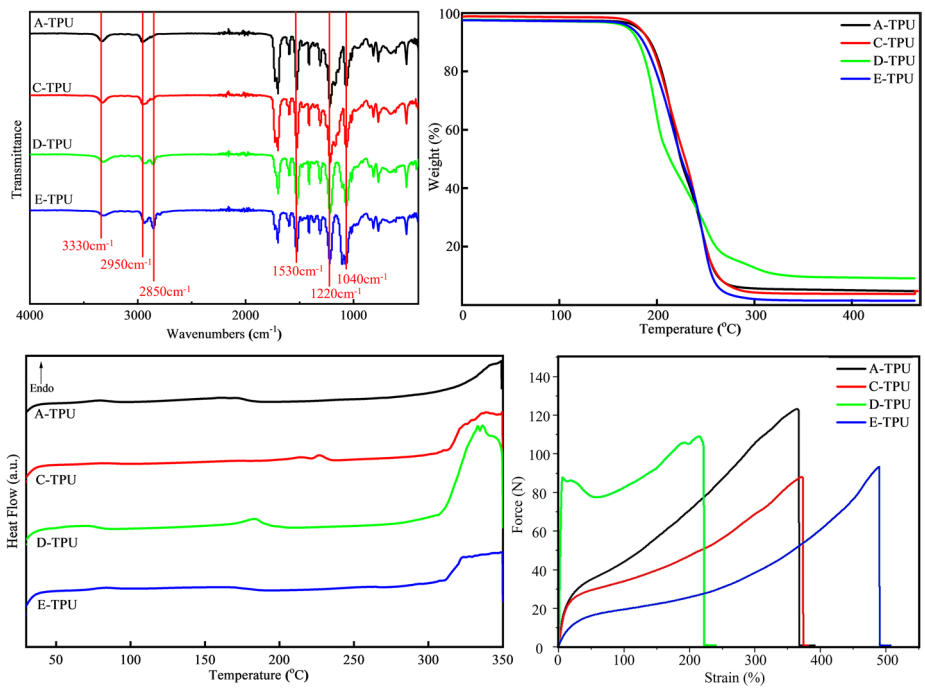
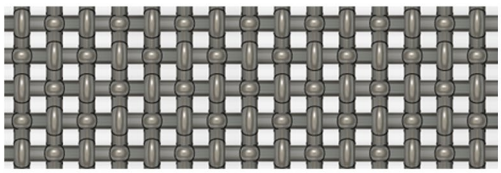
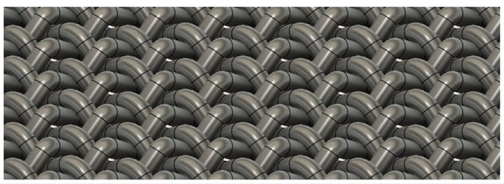


Fig. 3 FT-IR spectra, TGA curves, DSC analysis curves and Tensile strength of TPU filaments

Table 4 Modeling image of plain weave structure

Veiw	Modeling image
Top	
Front	
Side	
Perspective	

Surface morphologies of printed plain weave structures with modeling line thicknesses of 0.6 mm and 0.8 mm were compared using both Material Extrusion (MEX) and Conveyor Material Extrusion (CMEX) methods, as detailed in Tables 5 and 6.

Table 5 Surface image of 0.6 mm thickness Plain weave depending on gap size and printing angle

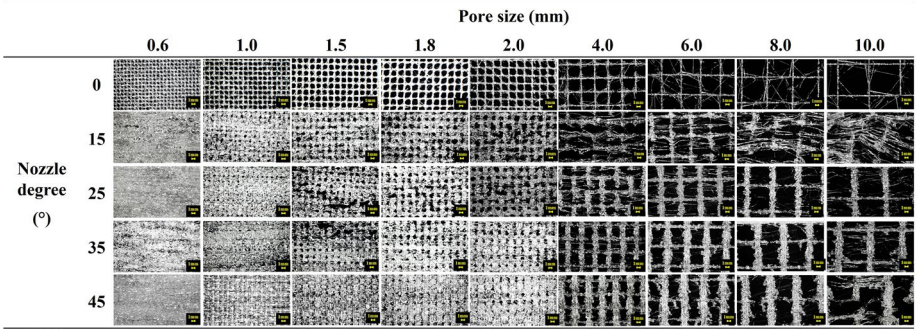
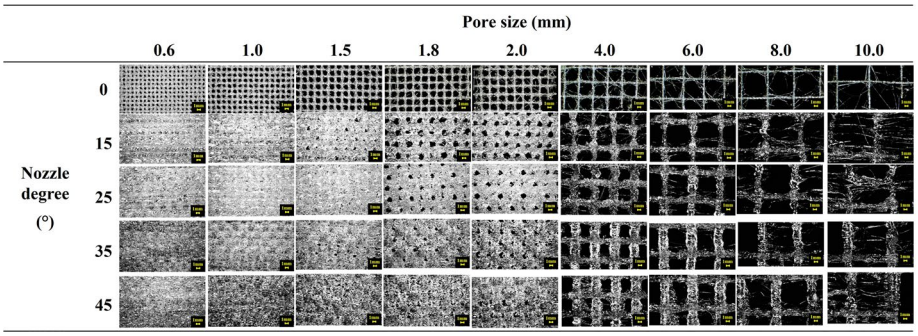


Table 6 Surface image of 0.8 mm thickness Plain weave depending on gap size and printing angle



MEX-printed samples accurately maintained the modeled line thickness and effectively represented the 0.6 mm gaps. In contrast, CMEX-printed samples exhibited increased line thickness and blurring as the printing angle increased, attributed to enhanced adhesion between the printed lines and the print bed. This resulted in lines spreading and becoming blurred, leading to no gaps being formed at a 0.6 mm gap size for both 0.6 mm and 0.8 mm modeling lines.

For plain weave structures, the optimal printing conditions were determined to be a 0.6 mm modeling line thickness with a 1.8 mm gap at a 15° printing angle, and a 0.8 mm modeling line thickness for gaps of 4.0 mm or more.

Plain knit

The plain knit structure is composed of loop forms; therefore, during modeling, pit point splines were utilized instead of basic line tools for sketching straight lines. When pit point splines are selected according to the loop size, custom splines are automatically formed between the chosen intervals, thereby accurately sketching the loop shape. The modeling images of the plain knit structures are presented in Fig. 4 and Table 7. For plain knits, line thicknesses of 0.6 mm, 0.8 mm, and 1.0 mm were employed during the modeling process. After sketching the loops, the modeling line thickness was increased by a factor of 1.5 to impart a three-dimensional appearance.

Surface analysis of plain knit outputs with modeling line thicknesses of 0.8 mm and 1.0 mm revealed that plain knits with a 0.6 mm modeling line did not produce successful prints using either MEX or CMEX methods due to the transition from two-dimensional to three-dimensional forms during the sketching process. As detailed in Table 8. MEX-printed samples with 0.8 mm modeling lines maintained the intended line thickness and accurately represented the 0.6 mm gaps, while CMEX-printed samples showed increased line thickness and blurring with higher printing angles (Table 8).

The optimal printing condition for plain knits was established as a 0.8 mm modeling line thickness with a 1.8 mm gap at a 15° printing angle.

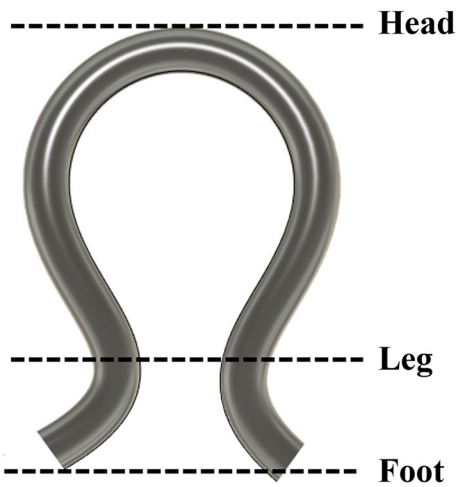


Fig. 4 One unit of the loop

Table 7 Modeling image of plain knit structure

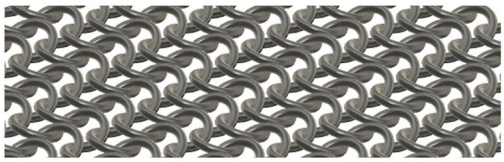
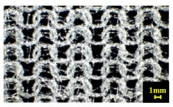
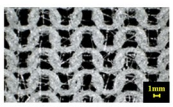
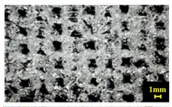
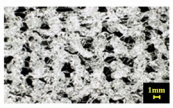
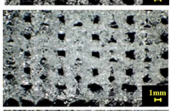
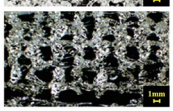
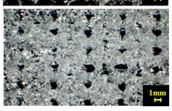
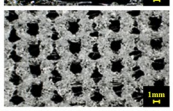
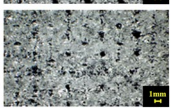
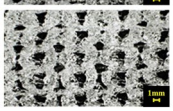
Veiw	Modeling image
Top	
Front	
Side	
Perspective	

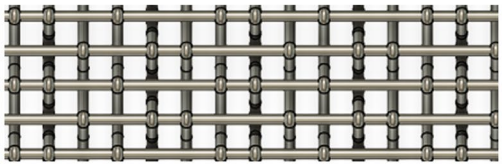
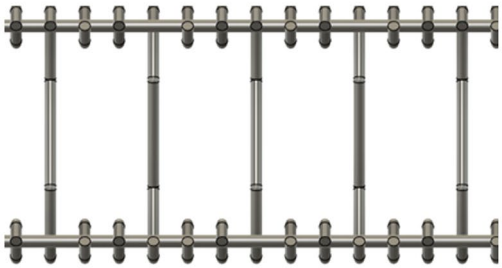
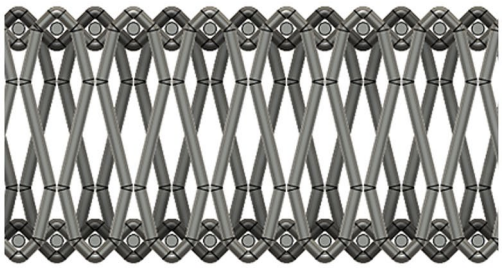
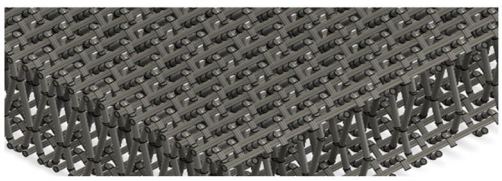
Table 8 Surface image of plain weave depending on thickness and printing angle

		Modeling line thickness (mm)		
		0.6	0.8	1.0
Nozzle degree (°)	0	-		
	15	-		
	25	-		
	35	-		
	45	-		

Spacer fabric

The modeling images of the spacer fabric structures are presented in Table 9. Spacer fabric is a three-dimensional textile structure, and modeling was conducted by maintaining the optimal line thickness for plain knit at 0.6 mm while incrementally increasing the gap size to 0.6 mm, 1.0 mm, 1.2 mm, 1.5 mm, and 1.8 mm. The spacer fabric sketches incorporated straight lines, zigzag lines typical of plain knits, and "figure-eight spacers"

Table 9 Modeling image of spacer fabric structure

Veiw	Modeling image
Top	
Front	
Side	
Perspective	

interweaving the surface and back straight lines. The height of the spacers, excluding the surface and back, was modeled to be 10 mm.

Surface analysis of spacer fabric structures revealed that, for MEX-printed outputs, samples without gaps exhibited minimal surface blurring, accurately reflecting the intended design. However, increasing the gap size led to surface blurring starting at a 1.2 mm gap. Conversely, CMEX-printed spacer fabrics without gaps showed significant surface blurring, obscuring the fabric structure as detailed in Tables 10, 11 and 12.

The optimal printing conditions for spacer fabric structures were determined to be a 1.2 mm gap and a 45° printing angle, where back thickness did not increase, surface blurring was absent, and uniform gaps were observed. For outputs requiring larger gaps of 4.0 mm or more, a modeling line thickness of 0.8 mm was recommended to maintain print quality.

Physical properties evaluation

The physical properties of the 3D-printed textile structures—plain weave, plain knit, and spacer fabric—were meticulously evaluated to assess their durability

Table 10 Surface image of spacer fabric depending on thickness and printing angle

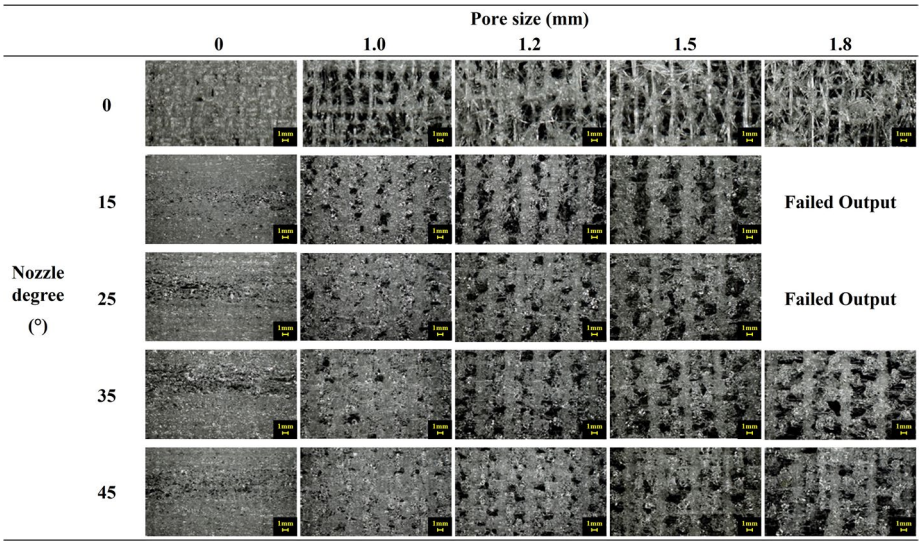
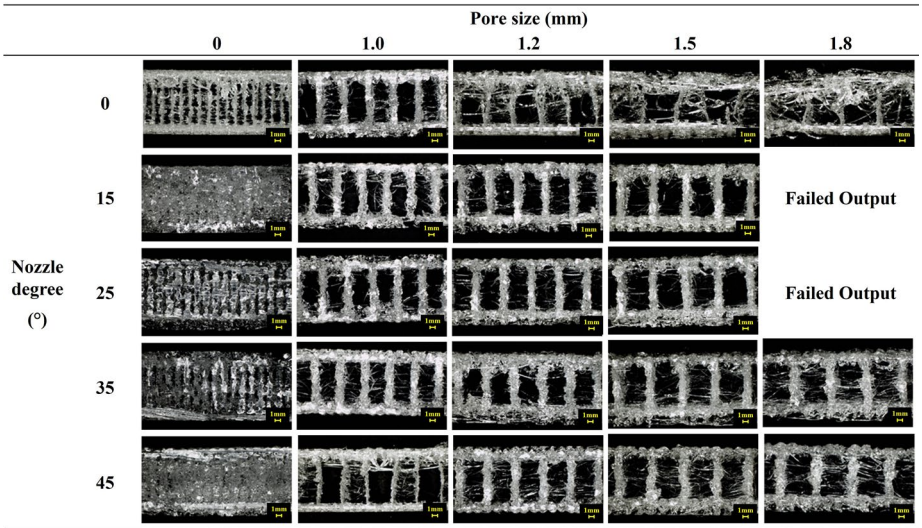


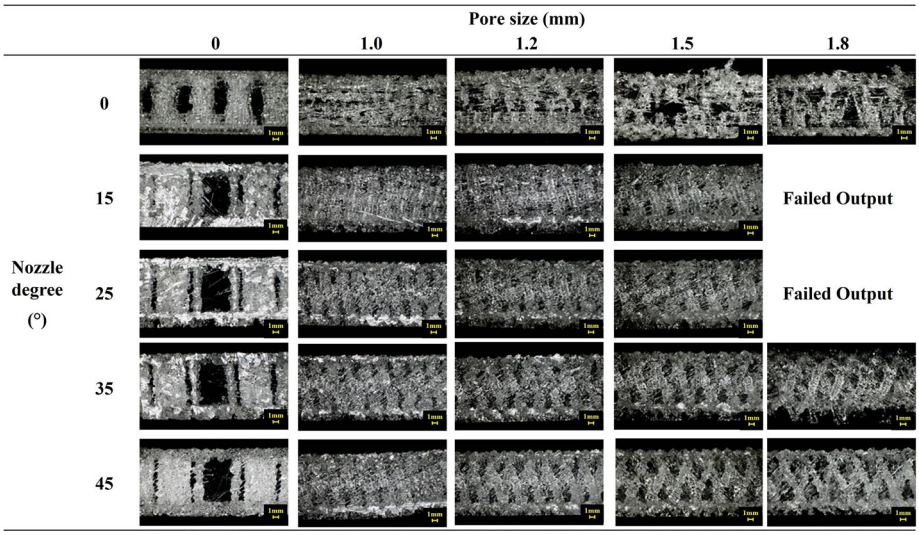
Table 11 Front image of spacer fabric depending on thickness and printing angle



and suitability for apparel applications (Fig. 5). The evaluations focused on tensile strength, flexural strength, tear strength, and stiffness, with results detailed below.

It is important to note that this tensile strength evaluation focuses on the 3D-printed textile structures themselves, distinct from the raw filament testing, in order to assess the mechanical behavior arising from their specific geometries and printing conditions. The tensile strength of the printed textile structures was assessed to determine the durability differences among plain weave, plain knit, and spacer fabric structures. Spacer fabrics exhibited the highest tensile strength, followed by plain knits and plain weaves. Specifically, spacer fabrics achieved a tensile strength of 0.069 N/mm^2 and an elongation rate of approximately 200%, which is six times higher than that of plain weave and plain knit structures. This finding confirms that as

Table 12 Side image of spacer fabric depending on thickness and printing angle



thickness increases, both elongation and tensile strength improve, aligning with the study by Nakajima et al. (2020), which demonstrated that increased thickness correlates with higher load-bearing capacity. Plain knits, characterized by loop forms with greater spacing between lines, demonstrated superior tensile strength and elongation compared to plain weaves, which have more interwoven lines. The enhanced tensile properties of spacer fabrics are attributed to their three-dimensional interwoven structure and increased TPU usage, which provide greater structural integrity. However, despite their high tensile strength, spacer fabrics may be prone to tearing under significant loads, suggesting their suitability for decorative rather than load-bearing applications.

Bending strength evaluations were conducted to understand the bending characteristics of the different textile structures. Spacer fabrics exhibited the lowest maximum bending displacement (11.80 mm), followed by plain weaves (18.66 mm), and plain knits (23.32 mm). Plain weaves and plain knits showed minimal flexural strength due to their thinness, making them easily bendable under minimal force, and thus, their flexural strength measurements were often unmeasurable. In contrast, spacer fabrics demonstrated measurable flexural strength (0.035 N/mm^2) and a higher maximum load capacity (0.056 N/mm^2). This is because flexural strength is influenced by material thickness (Kim & Lee, 2020). attributed to their increased thickness and robust structure. These results indicate that spacer fabrics are suitable for applications requiring impact absorption and structural integrity, such as shoe uppers, whereas plain weaves and knits are more appropriate for decorative or less load-bearing garments.

Tear strength testing further differentiated the structures. Spacer fabrics achieved the highest tear strength, followed by plain weaves and then plain knits. All specimens exceeded the recommended quality standard of 13 N for outer garments (Korea Consumer Agency, 2016), with spacer fabrics surpassing 40 N. The superior tear strength of spacer fabrics is attributed to their three-dimensional interlaced structure positioned

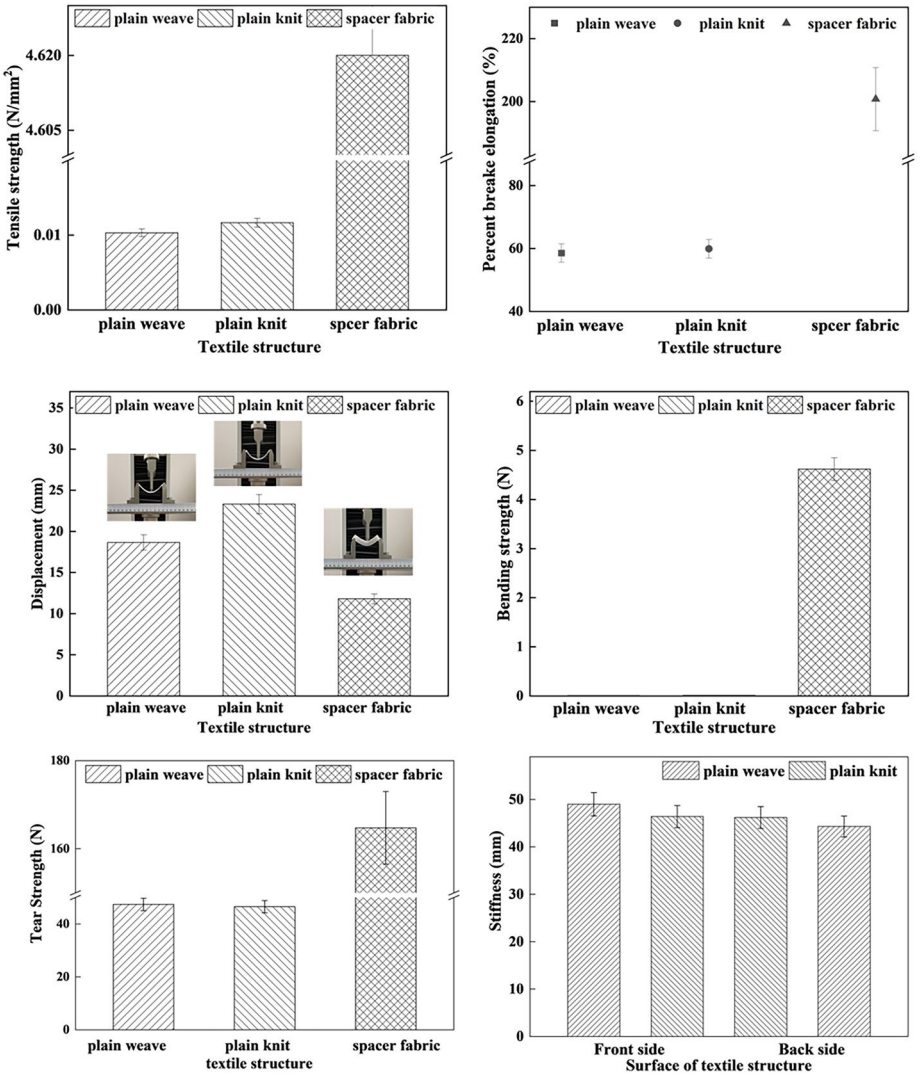
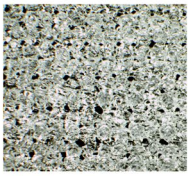
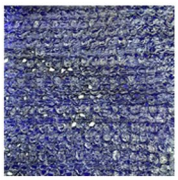
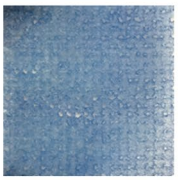


Fig. 5 Tensile properties(Tensile strength and Percent break elongation), Bending strength(Displacement and Bending strength), Tear strength and Stiffness of Three different 3D printing textiles using TPU filaments

between two layers of fabric, unlike conventional woven or knitted textiles, along with a higher TPU content, both of which enhance their resistance to tearing. Although the high tear strength of spacer fabrics reflects excellent durability, it may also imply a tendency for abrupt rupture under excessive load, suggesting that such structures may be more appropriate for decorative elements rather than load-bearing components in apparel.

Stiffness measurements were conducted for plain weave and plain knit structures spacer fabrics were excluded due to the confounding influence of spacer thickness. The results indicated that the back sides of both plain weaves and plain knits were slightly stiffer than their surfaces. Additionally, plain knits exhibited higher stiffness compared to plain weaves, correlating with the increased line thickness (0.8 mm) used in their modeling. This increased stiffness in plain knits suggests that these structures can be

Table 13 Color of TPU textile structure according to dye

	Undyed	Dyed with disperse dyes	Dyed with cationic dyes	Dyed with acid dyes
Sample				

tailored to produce garments that balance rigidity and flexibility, making them ideal for structured elements such as collars, cuffs, and ruffles without being overly rigid.

Evaluation of dyeing properties

To assess the dyeing potential of the 3D-printed TPU textile structures, dye affinity was evaluated using three types of dyes: disperse dye (Blue), basic dye, and acidic dye. Among the three printed plain weave structures with the smallest gaps, the plain weave was selected for this assessment. The plain weave structure was selected for dyeing tests because it represents the most fundamental textile architecture with lower areal weight, enabling consistent evaluation of dye uptake efficiency using identical TPU material while minimizing dye consumption and processing time. Presents images of the dyed textile structures, as shown in Table 13. This choice ensured a consistent evaluation on specimens printed with the same TPU material and comparable processing conditions. Although nozzle angles varied slightly to enable specific structural geometries, the use of identical nozzle temperature and speed allowed us to treat these as equivalent printing conditions for assessing dyeing performance.

The results showed that both basic and acidic dyes exhibited minimal to no coloration on the TPU structures. This outcome is consistent with the nature of these dyes: basic dyes are typically used for protein fibers such as wool and silk, as well as acrylic fibers, while acidic dyes are suited for amine-containing fibers like nylon and wool. In contrast, disperse dyes, which are intended for hydrophobic fibers such as acetate, nylon, and polyester, demonstrated the most effective dyeing on the TPU structures. This effectiveness is attributed to the hydrophobic nature of polyurethane fibers and their compatibility with disperse dyes, which are known for their good color fastness when applied under high-temperature and high-pressure conditions, as noted in previous studies (KOTITI Testing & Research Institute, 2025).

The dye uptake rate of disperse dyes within the 3D-printed TPU textile structures was measured to evaluate the degree of dye penetration. Figure 6 is absorbance measurements before and after dyeing using a 1% aqueous solution of a 3% (o.w.f) disperse dye solution. The maximum absorbance for the blue disperse dye was observed at a wavelength of 733.5 nm. The absorbance values decreased from 0.14 before dyeing to 0.11 after dyeing, indicating a 24% reduction in absorbance post-dyeing. This lower dye uptake rate is likely due to the high-temperature and high-pressure conditions typically required for effective disperse dyeing, which were not fully met in this study. It is

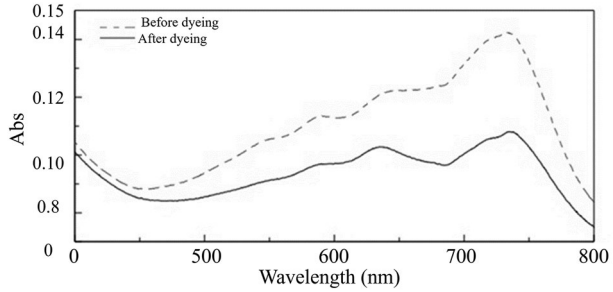


Fig. 6 UV–Vis absorption spectra of the dye before and after dyeing

Table 14 Colorimetric data of dyed fabric according to washing and dry cleaning

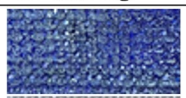
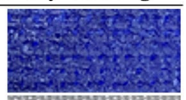
	K/S (λ733.5nm)	L	a	b
Undyed	0.4	23.44	9.73	-18.91
after washing	0.31	24.19	8.8	-18.6
after dry cleaning	0.9	25.89	19.16	-36.33

anticipated that increasing the dyeing temperature would enhance the dye uptake rate, thereby improving the overall dyeing efficiency of the TPU structures.

To evaluate the morphological changes and dye fixation of the TPU textile structures dyed with disperse dyes, K/S values and colorimetric parameters (L, a, b) were measured before and after washing and dry cleaning. Table 14 presents the results, indicating no significant morphological deformation after washing or dry cleaning, which is attributed to the high durability of TPU material and the high-temperature nozzle extrusion process used in 3D printing. The K/S values showed minimal changes before and after washing, suggesting stable color retention. However, after dry cleaning, there was a noticeable shift in colorimetric values (a and b directions), resulting in a more yellow and green hue in the blue-dyed samples. This color change is likely due to the organic solvents used during dry cleaning, which can induce yellowing in TPU materials.

The wash fastness of the dyed TPU textile structures was further evaluated using attached padding tests to assess color fading and staining. Table 15 details the wash fastness results, where the color fading fastness after washing and dry cleaning was rated at 4.5 and 3.5, respectively. These ratings meet the recommended quality standards for outer garments, which require a color fading fastness of grade 4. However, staining fastness results varied depending on the fiber type of the attached padding. While acetate, polyester, acrylic, and wool blends achieved a staining fastness rating of grade 3, nylon exhibited a significantly lower rating of 1.5, indicating a high propensity for color bleeding. Consequently, it is recommended that TPU textile structures dyed with disperse dyes be washed separately or not mixed with nylon-containing fabrics to prevent dye transfer and maintain color integrity.

Table 15 Color fastness rating grade of dyed fabric after washing and dry-cleaning

		After washing		After dry-cleaning	
Fade		4.5		3.5	
Stain	Acetate	3		3	
	Cotton	3.5		3.5	
	Nylon	1.5		2	
	Polyester	4		2.5	
	Acrylic	4.5		4.5	
	Rayon	3		3	

Conclusions

The exploration of TPU-based 3D-printed textiles using CMEX and MEX technologies has demonstrated the potential for creating continuous, flexible textile structures suitable for apparel applications. By optimizing print parameters and selecting TPU filaments based on their thermal and physical properties, this study provides foundational data for designing diverse textile architectures such as plain weave, plain knit, and spacer fabrics.

For plain weave structures, optimal surface morphology was achieved at a 15° printing angle with a line thickness of 0.6 mm and a gap of 1.8 mm. Flat fabric structures exhibited the most stable surface at a 25° printing angle with a 0.8 mm line thickness, while spacer fabrics demonstrated the highest stability at a 45° printing angle with a gap size of 1.2 mm. These findings suggest that different printing angles must be controlled for each printed structure, highlighting the necessity of precise modeling and parameter control to achieve the desired surface quality and structural stability when utilizing CMEX and MEX 3D printing.

Physical property evaluations revealed that spacer fabrics offer high tensile and tear strength, along with superior flexural stability, making them suitable for structural or impact-absorbing applications. In contrast, plain weaves and knits are more appropriate for flexible or decorative garment components. Dyeing assessments confirmed that TPU textiles respond best to disperse dyes, with stable color retention after washing and slight changes following dry cleaning. Washing and staining fastness results suggest that TPU textiles, particularly when combined with nylon-based fabrics, should be washed separately to maintain their color integrity.

In conclusion, this study highlights the feasibility of textile manufacturing using 3D printing technology, demonstrating the potential for integrating TPU-based materials

into continuous textile structures and presenting new opportunities for sustainable fashion. However, despite these advancements, 3D-printed textiles still face technical limitations in their application as garment materials. To further advance the development of 3D-printed textiles, additional research is needed to produce thinner structures that more closely resemble conventional fabrics, thereby enhancing their suitability for practical textile applications.

Acknowledgements
This research was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (No. NRF-2021R1I1A3057957)

Author contributions
Minseo Kim contributed to conceptualization(lead), data curation(lead), writing—original draft preparation(lead), writing—reviewing and editing(equal); Sohee Lee contributed to funding acquisition (lead), supervision (lead), writing—reviewing and editing(equal).

Data availability
The data that support the findings of this study are available from the corresponding authors upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The authors declare that there are no competing interests regarding the publication of this manuscript.

Received: 27 May 2025 Accepted: 19 October 2025
Published online: 20 November 2025

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